Potassium permanganate - $KMnO_4$

Preparation:

Potassium permanganate is prepared from pyrolusite (MnO_2) ore. The preparation involves the following steps.

(i) Conversion of MnO2 to potassium manganate:
Powdered ore is fused with KOH in the presence of air or oxidising agents like KNO_3 or $KClO_3$. A green coloured potassium manganate is formed.

$$2MnO_2 + 2KOH + O_2 \longrightarrow \underset{\substack{\textit{potassium manganate}\\(\textit{Green})}}{K_2MnO_4} + 2H_2O$$

(ii) Oxidation of potassium manganate to potassium permanganate:
Potassium manganate thus obtained can be oxidised in two ways , either by chemical oxidation or electrolytic oxidation.

Chemical oxidation:

In this method potassium manganate is treated with ozone (O_3) or chlorine to get potassium permanganate.

$$2MnO_4^{2-} + O_3 + H_2O \longrightarrow 2MnO_4^{-} + 2OH^{-} + O_2$$

$$2MnO_4^{2-} + Cl_2 \longrightarrow 2MnO_4^{-} + 2Cl^{-}$$

Electrolytic oxidation

In this method aqueous solution of potassium manganate is electrolyzed in the presence of little alkali.

$$K_2MnO_4 \rightleftharpoons 2K^{+} + MnO_4^{2-}$$

$$H_2O \rightleftharpoons H^{+} + OH^{-}$$

Manganate ions are converted into permanganate ions at anode.

$$\underset{\textit{Green}}{MnO_4^{2-}} \rightleftharpoons \underset{\textit{purple}}{MnO_4^{-}} + e^{-}$$

$H_2$is liberated at the cathode.

$$2H^{+} + 2e^{-} \longrightarrow H_2 \uparrow$$

The purple coloured solution is concentrated by evaporation and forms crystals of potassium permanganate on cooling.

Physical properties:

Potassium permanganate exists in the form of dark purple crystals which melts at 513 K. It is sparingly soluble in cold water but, fairly soluble in hot water.

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